

1. Total Number of Patients

Write an SQL query to find the total number of patients across all hospitals.

ANS: SELECT SUM("Patients Count") AS "Total Patients"

FROM hospital_data;

9347

2. Average Number of Doctors per Hospital

Retrieve the average count of doctors available in each hospital.

ANS: SELECT "Hospital Name",

AVG("Doctors Count") AS "Average Doctors"

FROM hospital_data

GROUP BY "Hospital Name";

"City Hospital"	24.700000000000000
"Healing Touch"	25.6428571428571429
"Global Medicare"	28.727272727272723
"Fortis Care"	24.545454545454545
"Sunrise Medical"	28.625000000000000
"Green Valley Hospital"	28.875000000000000
"Heritage Hospital"	24.666666666666667
"Metro Hospital"	27.45454545454545
"Apollo Health"	23.36363636363636
"Wellness Clinic"	34.000000000000000

3. Top 3 Departments with the Highest Number of Patients

Find the top 3 hospital departments that have the highest number of patients.

ANS: SELECT "Department",

SUM("Patients Count") AS "Total Patients"

FROM hospital_data

GROUP BY "Department"

ORDER BY "Total Patients" DESC

LIMIT 3;

Department	Total Patients
Urology	1766
Neurology	1229
ENT	1064

4. Hospital with the Maximum Medical Expenses

Identify the hospital that recorded the highest medical expenses.

```
ANS: SELECT "Hospital Name",
        MAX("Medical Expenses") AS "Maximum Medical Expenses"
FROM hospital_data
GROUP BY "Hospital Name"
ORDER BY "Maximum Medical Expenses" DESC
LIMIT 1;
```

Hospital Name	Maximum Medical Expenses
HealingTouch	49955.41

5. Daily Average Medical Expenses

Calculate the average medical expenses per day for each hospital.

```
ANS: SELECT
    "Hospital Name",
    ROUND(
        AVG(
            "Medical Expenses" /
            (
                (
                    CASE
                        WHEN "Discharge Date" LIKE '%/%'
                        THEN TO_DATE("Discharge Date", 'DD/MM/YYYY')
                        ELSE TO_DATE("Discharge Date", 'DD-MM-YYYY')
```

```

END
-
CASE
    WHEN "Admission Date" LIKE '%/%'
    THEN TO_DATE("Admission Date",'DD/MM/YYYY')
    ELSE TO_DATE("Admission Date",'DD-MM-YYYY')
END
) + 1
)
),
2
) AS "Daily Average Expenses"
FROM hospital_data
GROUP BY "Hospital Name";

```

Hospital Name	DailyAverageExpenses
City Hospital	6298.17
Healing Touch	3830.7
Global Medicare	5585.86
Fortis Care	6388.53
Sunrise Medical	5230.81
GreenValleyHospital	4820.35
Heritage Hospital	4089.24
Metro Hospital	5230.65
Apollo Health	4034.79
Wellness Clinic	4135.77

6. Longest Hospital Stay

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

ANS: SELECT

```

"Hospital Name",
"Department",
"Admission Date",

```

```

"Discharge Date",
(
CASE
    WHEN "Discharge Date" LIKE '%/%'
    THEN TO_DATE("Discharge Date",'DD/MM/YYYY')
    ELSE TO_DATE("Discharge Date",'DD-MM-YYYY')
END
-
CASE
    WHEN "Admission Date" LIKE '%/%'
    THEN TO_DATE("Admission Date",'DD/MM/YYYY')
    ELSE TO_DATE("Admission Date",'DD-MM-YYYY')
END
) AS "Stay Days"
FROM hospital_data
ORDER BY "Stay Days" DESC
LIMIT 1;

```

HospitalName	Department	AdmissionDate	DischargeDate	StayDays
Apollo Health	ENT	6/4/2023	21-04-2023	15

7. Total Patients Treated Per City

Count the total number of patients treated in each city.

```

ANS: SELECT "Location",
    SUM("Patients Count") AS "Total Patients"
FROM hospital_data
GROUP BY "Location"
ORDER BY "Total Patients" DESC;

```

Location	TotalPatients
Jaipur	1505
Ahmedabad	1467
Lucknow	1264
Hyderabad	1261
Bangalore	955
Chennai	840
Kolkata	596
Pune	555
Mumbai	483
Delhi	421

8. Average Length of Stay Per Department

Calculate the average number of days patients spend in each department.

```

ANS: SELECT "Department",
    ROUND(
        AVG(
            CASE
                WHEN "Discharge Date" LIKE '%/%'
                THEN TO_DATE("Discharge Date", 'DD/MM/YYYY')
                ELSE TO_DATE("Discharge Date", 'DD-MM-YYYY')
            END
        - CASE
                WHEN "Admission Date" LIKE '%/%'
                THEN TO_DATE("Admission Date", 'DD/MM/YYYY')
                ELSE TO_DATE("Admission Date", 'DD-MM-YYYY')
            END
        ),
    ) AS "Average Stay Days"
FROM hospital_data
GROUP BY "Department";

```

Department	AverageStayDays
Dermatology	5.6
Oncology	8.11
ENT	8.08
Neurology	9.25
Gynecology	7.67
Pediatrics	9.11
Orthopedics	7.14
Cardiology	6.86
Urology	8.72
General Medicine	7.43

9. Identify the Department with the Lowest Number of Patients

Find the department with the least number of patients.

```
ANS: SELECT "Department",
        SUM("Patients Count") AS "Total Patients"
FROM hospital_data
GROUP BY "Department"
ORDER BY "Total Patients" ASC
LIMIT 1;
```

Department	Total Patients
Cardiology	544

10. Monthly Medical Expenses Report.

Group the data by month and calculate the total medical expenses for each month.

```
ANS: SELECT
    TO_CHAR(
        CASE
            WHEN "Admission Date" LIKE '%/%'
            THEN TO_DATE("Admission Date", 'DD/MM/YYYY')
            ELSE TO_DATE("Admission Date", 'DD-MM-YYYY')
        END,
```

```

        'Month'
    ) AS "Month",
    SUM("Medical Expenses") AS "Total Medical Expenses"
FROM hospital_data
GROUP BY
    TO_CHAR(
        CASE
            WHEN "Admission Date" LIKE '%/%'
            THEN TO_DATE("Admission Date", 'DD/MM/YYYY')
            ELSE TO_DATE("Admission Date", 'DD-MM-YYYY')
        END,
        'Month'
    ),
    EXTRACT(
        MONTH FROM
        CASE
            WHEN "Admission Date" LIKE '%/%'
            THEN TO_DATE("Admission Date", 'DD/MM/YYYY')
            ELSE TO_DATE("Admission Date", 'DD-MM-YYYY')
        END
    )
ORDER BY
    EXTRACT(
        MONTH FROM
        CASE
            WHEN "Admission Date" LIKE '%/%'
            THEN TO_DATE("Admission Date", 'DD/MM/YYYY')
            ELSE TO_DATE("Admission Date", 'DD-MM-YYYY')
        END);

```

Month	TotalMedicalExpenses
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January	173971.54
February	301722.72
March	199247.42
April	88995.93
May	222986.72
June	165926.36
July	211527.13
August	181039.55
September	341284.23
October	158450.68
November	334370.33
December	337788.45
